

Representation-Relative Grokking

Grokking as Representation Discovery in Neural Mathematical Reasoning

David Vesterlund

Affiliation: [to be finalized]

Preprint Draft v0.1 - September 2026

Status: hypothesis paper with preliminary motivating evidence. M04/M05 results are treated as preliminary and do not by themselves establish the general hypothesis. Several causal, temporal, scaling, and cross-task tests are explicitly proposed below.

Abstract

Grokking is usually described as delayed generalization after a neural network has already fit its training set. We propose a more specific hypothesis for structured mathematical tasks: grokking is often a transition in the representation-computation pair used by the learner. Under the Representation-Relative Grokking Hypothesis, a network can initially fit examples in an easily accessible but poorly generalizing representation, while prolonged optimization can induce a transition toward a symmetry-adapted representation in which a shared mathematical rule becomes simpler to implement. Crucially, such a representation need not be unique. We distinguish the cost of discovering a representation from the complexity of solving the task once the representation is available, yielding an accessibility-efficiency frontier across possible internal solutions. Preliminary evidence comes from two transformers trained on the same discrete-logarithm task over the multiplicative group of F_{113} . A 427k-parameter model (M04) groks later and admits a sparse divisor-aligned spectral decision rule, whereas a 657k-parameter model (M05) groks substantially earlier while using a denser and qualitatively different spectral representation. This motivates the hypothesis that model capacity changes not only whether grokking occurs, but which mathematical representation is selected. We formalize the hypothesis, derive falsifiable predictions, and propose a study program based on explicit representation interventions, prime-order controls, checkpoint-level mechanistic analysis, causal ablations, capacity scaling, and cross-task convergence between multiplicative DLP and elliptic-curve DLP. The central claim is not that mathematical difficulty disappears under a change of coordinates, but that neural learnability depends on the joint cost of discovering a useful representation and computing within it.

Keywords: grokking; mechanistic interpretability; representation learning; mathematical reasoning; discrete logarithm; Fourier analysis; group representations; neural algorithms

1. Introduction: Mathematical Difficulty Is Representation-Relative

Grokking, first highlighted on small algorithmic datasets, refers to the delayed emergence of test generalization long after a model has already achieved near-perfect training performance [1]. Subsequent mechanistic work on modular addition showed that the apparently abrupt transition can be decomposed into gradual circuit formation and cleanup, with Fourier-structured computation emerging before the visible accuracy jump [2]. Other work has emphasized circuit efficiency [5] and transitions from lazy to feature-learning dynamics [6]. These accounts explain important aspects of delayed generalization, but they leave open a broader question that is especially salient for mathematical tasks: what if the decisive event is not only the formation of a better circuit, but the discovery of a better mathematical representation in which that circuit becomes simple?

Mathematics is full of examples where a problem changes character under a change of coordinates. Translation becomes phase rotation in a Fourier basis; finite abelian groups decompose into character coordinates; Chinese-remainder coordinates split arithmetic over coprime factors; projective and quotient representations remove symmetries that are redundant for a computation. The computational object is the same, yet the complexity of the operation can change sharply with the representation.

We therefore propose to treat mathematical grokking as a representation-sensitive phenomenon. The network is not merely choosing between memorization and an algorithm. It is choosing, implicitly through gradient-based optimization, among representation-computation pairs. This paper develops that hypothesis and uses preliminary DLP results as a motivating case rather than as definitive proof.

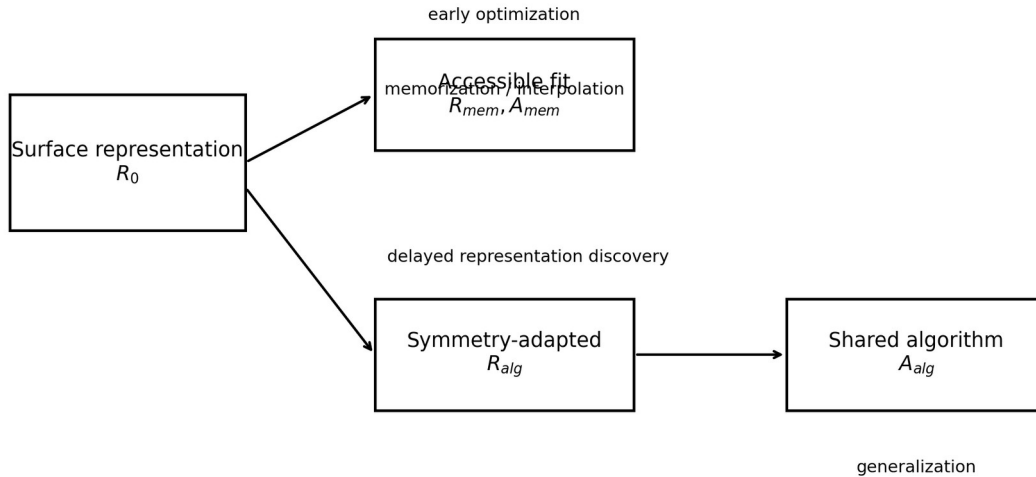


Figure 1. Conceptual model: grokking as delayed discovery of a representation-computation pair that supports a shared generalizing rule.

2. The Representation-Relative Grokking Hypothesis

Let $f: X \rightarrow Y$ be a structured mathematical task, and let $\rho_0: X \rightarrow Z_0$ denote the surface representation in which training examples are presented. Suppose there exists an alternative representation $R: X \rightarrow Z_R$ and a solver g_R such that the target can be implemented as:

$$f(x) = g_R(R(x)).$$

The central hypothesis is:

Representation-Relative Grokking Hypothesis. For structured mathematical tasks, delayed

generalization often occurs when optimization discovers an internal representation in which the target mapping can be implemented by a more general, symmetry-aligned computation than is available in the original surface representation. Multiple such representations may exist, and the representation that emerges depends jointly on mathematical structure, architecture, capacity, regularization, data distribution, and optimization dynamics.

A stronger but still explicitly testable form is:

$$(R_mem, A_mem) \rightarrow (R_alg, A_alg),$$

where R_mem supports example-specific fitting and R_alg supports a shared rule that generalizes. The claim is deliberately weaker than "all grokking is representation discovery". The intended domain is a broad but falsifiable class of structured mathematical tasks.

3. Representation Discovery versus Solver Complexity

The hypothesis separates two quantities that are often conflated.

3.1 Representation discovery cost

Let $D(R \mid \rho_0, A, O, D_data)$ denote the difficulty, under architecture A , optimizer O , and data distribution D_data , of reaching a representation functionally equivalent to R from the surface encoding ρ_0 . Operational proxies include the optimization steps, processed examples, or compute required before a stable representation-alignment score crosses a preregistered threshold.

3.2 Solver complexity

Let $S_A(f \mid R)$ denote the complexity of the computation required by architecture A to solve f once representation R is available. Candidate empirical proxies include minimal distilled circuit size, spectral participation ratio, causal spectral dimension, effective logit rank, description length, and the number of indispensable heads or subspaces.

3.3 A conceptual neural mathematical complexity

$$C_neural(f; \rho_0, A, O, D_data) = \min_R [D(R \mid \rho_0, A, O, D_data) + \lambda S_A(f \mid R)].$$

This expression is currently a conceptual functional rather than a universal scalar law: D and S do not yet share canonical units. Its purpose is to state the decomposition clearly. A mathematically elegant representation can have very low solver complexity but very high discovery cost, while a redundant representation can be easier for gradient descent to reach.

4. The Accessibility-Efficiency Frontier

If several symmetry-compatible representations solve the same task, optimization need not select the representation with the smallest final circuit. Instead, learned solutions can lie on a frontier trading off accessibility during training against efficiency after discovery. We call this the Representation Accessibility-Efficiency Frontier.

This differs from circuit-efficiency accounts of grokking [5] in one important respect. Circuit efficiency compares competing implementations in parameter space; our proposal introduces the representation itself as a latent choice variable. A circuit may be efficient only after a particular coordinate system has been discovered.

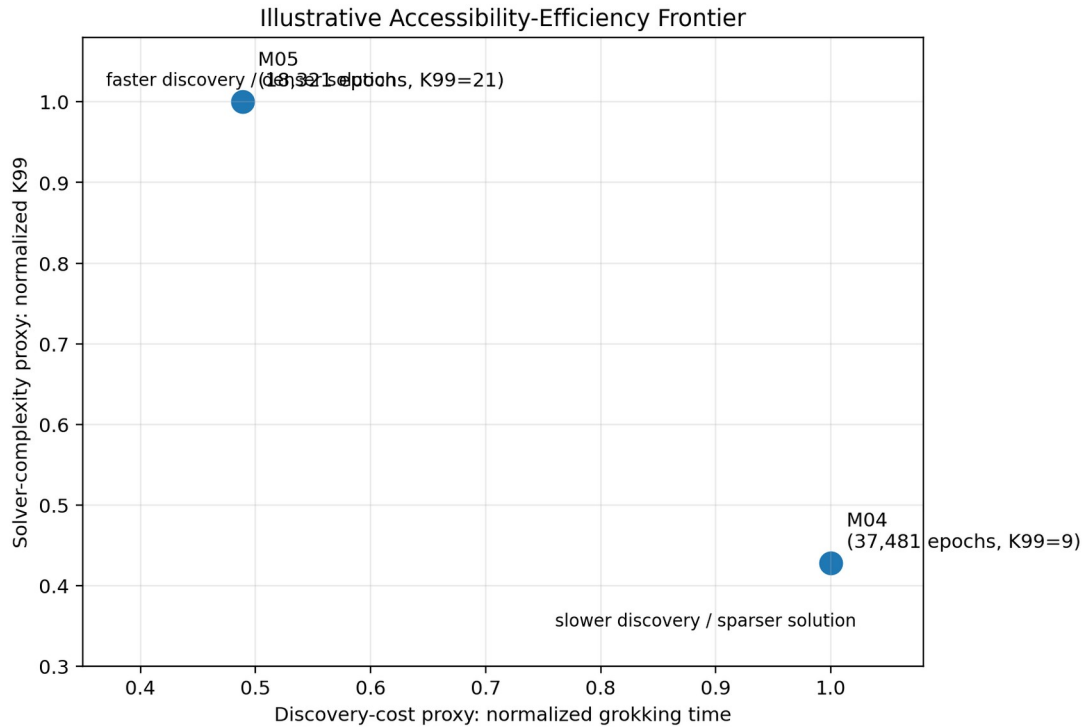


Figure 2. Illustrative proxy view of M04 and M05. Grokking time is used as a discovery-cost proxy and K99 as a solver-complexity proxy. The axes are not yet universal complexity measures.

5. Capacity-Dependent Representation Selection

Capacity-Dependent Representation Selection Hypothesis. Increasing model capacity changes not only whether a task can grok, but which internal representation is selected. Larger models may prefer a denser or less compressed representation if it is easier to discover, while smaller models may be pushed toward a sparser but harder-to-find representation.

This hypothesis predicts that model scale can change the algorithmic path even when task, data, and training objective are held fixed. It also predicts that a faster grokking model need not end with the simplest final circuit. This is a direct empirical question that can be tested across the existing M01-M16 scaling family.

6. Case Study I: M04 and M05

We use two grokked transformers on the same DLP task over F_{113}^* as a motivating case. The group order is $q = 112 = 2^4 * 7$. These observations are preliminary and are used here to generate hypotheses and experimental predictions, not to establish the general theory.

Property	M04	M05
Parameters	~427k	~657k
Architecture	2 layers; d_model 128; 4 heads	2 layers; d_model 160; 5 heads
Test accuracy	99.8%	99.1%
Reported grokking epoch	37,481	18,321

Property	M04	M05
Key frequencies for 100% closed-form argmax	9 (~5 unique)	21 (~11 unique)
Dominant mode	k=28, period 4	k=56, period 2
Other prominent modes	period 7 and 16 structure	period 14 structure
Closed-form argmax	100%	100%
Sine coefficients	near zero	near zero
Logit circulancy	approximately yes	no; large violations reported

Table 1. Preliminary M04/M05 comparison from the internal extraction report. Exact final values should be regenerated from frozen checkpoints before submission.

6.1 Same task, different spectral routes

For M04, the dominant modes align with short periods induced by divisors of 112: k=28 has period 4; k=16, 32, and 48 have period 7; and the canonical partner of k=91 is k=21, which has period 16. This is consistent with a CRT/Pohlig-Hellman-like factorized representation, although causal evidence is still required before making a mechanistic identity claim.

M05 emphasizes a different set. Its dominant k=56 mode has period 2, while k=8, 24, 88, and 104 each have period 14. Thus M05 also appears divisor-aligned, but along a denser route through the same cyclic group. The larger model groks earlier while using more frequencies. This pattern is consistent with, but does not prove, an accessibility-efficiency trade-off.

6.2 Important mechanistic caution

The current closed-form spectral surrogate should not yet be described as a complete extraction of the DLP computation. For M04, nine selected frequencies can reproduce 100% argmax decisions while explaining only about 14.6% of logit variance; the fit saturates around $R^2 \sim 0.157$ with more frequencies. Decision sufficiency is therefore not the same as full circuit equivalence. For M05, the reported non-circulant logit structure is additional evidence that its computation is not simply the identical M04 circuit with different coefficients. These facts strengthen the motivation for causal ablations, residual-logit tests, and checkpoint-level circuit reconstruction.

7. Experimental Tests of Representation Causality

The hypothesis is only interesting if it can be falsified. We therefore propose tests that move from correlation to temporal evidence and then to causal intervention.

H1 - Representation emergence precedes generalization. A candidate representation should become decodable before or during the grokking transition, not only after generalization is complete.

H2 - Explicit representation accelerates learning. Providing a candidate symmetry-adapted representation directly should reduce discovery time, sample complexity, or eliminate the delayed phase.

H3 - Causal ablation selectively harms generalization. Removing the representation-bearing subspace should damage OOD/test behavior more than rank-matched random ablations.

H4 - Representation quality predicts sample complexity. Representations with lower measured solver complexity should require fewer examples after discovery, conditional on accessibility.

H5 - Representation-aligned OOD generalization. A model that learned the underlying representation should generalize to unseen surface configurations that preserve the same latent structure.

H6 - Structural changes induce different mechanisms. Smooth composite-order and prime-order DLP should not be forced into the same short-period circuit if the relevant subgroup structure disappears.

H7 - Equivalent latent mathematics predicts convergence. Different surface tasks with the same latent algebra should develop more similar internal representations after grokking than before it.

7.1 Explicit CRT representation intervention

For y in F_{113}^* , define the algebraic projections $y_{16} = y^7 \bmod 113$ and $y_7 = y^{16} \bmod 113$. Because the group has order $112 = 16 * 7$, these projections expose order-16 and order-7 components without computing a discrete logarithm. We propose matched training on RAW, CRT-BOTH, CRT-16-ONLY, CRT-7-ONLY, RAW+CRT, and a scrambled-pair control, with identical underlying examples and training budgets.

A strong result would be that CRT-BOTH generalizes substantially earlier than RAW, while the single-component variants reach the expected information ceilings for full- x prediction and the scrambled control does not reproduce the acceleration. This would directly support the claim that an algebraically aligned representation reduces discovery cost rather than merely adding tokens.

8. Prime-Order versus Factorizable Groups

The current motivating task has exponent-group order 112, which factors as $16 * 7$. To determine whether the observed short-period representation is necessary for grokking or merely one convenient implementation, we propose a matched prime-order control.

Use the order-113 subgroup H of F_{227}^* , since $227 - 1 = 2 * 113$. Keep the M04 architecture, optimizer, training schedule, and vocabulary size matched as closely as possible. Compare both coverage-matched and sample-count-matched regimes, and include a Base-OOD split where generators are disjoint across train and test. In the prime-order subgroup, every non-trivial character has full period 113, so the period-4, period-7, and period-16 shortcut available at $q=112$ cannot exist.

The outcomes have different implications: failure at $r=113$ would support the view that smooth factorization is materially enabling; slower grokking would suggest factorization is an accelerator but not a prerequisite; similar grokking with a qualitatively new spectrum would strongly support representation adaptivity; and similar grokking with no detectable representational change would weaken the current mechanistic account.

9. Cross-Problem Representation Convergence

The strongest test of the hypothesis is not another DLP modulus. It is a pair of surface-different tasks that share latent mathematics. Variable-base multiplicative DLP and EI4/ECDLP are natural candidates:

$$\text{multiplicative DLP: } g^x = h,$$

$$\text{elliptic DLP: } [x]A = B.$$

In both cases, after choosing analysis-only cyclic coordinates, the latent relation is scalar arithmetic in a cyclic group. The prediction is not that the raw activations should match, but that grokked models should become more alignable to a common cyclic-coordinate control than their pre-grok checkpoints are. This can be measured with CKA, RSA, CCA, Procrustes alignment, subspace angles, and coordinate probes.

9.1 Pairing-mediated representation experiment

A later experiment can use an elliptic-curve pairing representation. For suitable torsion points P , $Q=[x]P$, and an independent point T , the Weil pairing satisfies $e_r(Q,T) = e_r(P,T)^x$. This converts the scalar relation into a multiplicative-group DLP in the pairing image. Comparing direct EC-point inputs with pairing-derived inputs provides a particularly sharp representation intervention: the mathematical unknown x is unchanged, while the surface representation and computational geometry are altered.

10. Implications for Neural Mathematical Reasoning

If supported, the hypothesis suggests that mathematical capability in neural networks should be studied in terms of representation search rather than only function approximation. A network can appear weak on a task because the current coordinates make the relevant rule difficult to discover, not because the underlying rule is beyond its capacity once expressed appropriately.

This suggests a research program around representation-native mathematical models: Fourier/character-native models for cyclic structure, p-adic representations for valuation and congruence structure, tropical or Newton-polygon representations for piecewise-linearized algebra, projective/Kummer representations for elliptic-curve quotients, generating-function representations for recurrences, and representation-theoretic bases for non-abelian group tasks. The critical experimental question is not whether a representation is mathematically elegant, but whether it changes discovery cost, solver complexity, and OOD generalization in predictable ways.

11. Connection to Representation Scheduling in Computation

The same meta-problem appears in algorithm design outside neural networks. In quantum algorithm research, one may choose among mathematical, fermionic, operator, circuit, and variational representations to minimize qubits, depth, two-qubit gates, block-encoding normalization, or physical error burden. In neural learning, one may choose among representations to minimize examples, optimization time, parameter requirements, or learned-circuit complexity.

$$R^* = \operatorname{argmin}_R C(\text{computation} \mid R).$$

The cost functional differs by computational substrate, but the structural question is shared: which representation makes the desired computation easiest under the available machine? This paper focuses on neural grokking, but the broader perspective motivates future work on dynamic representation selection and scheduling.

12. Limitations and Falsification Criteria

The hypothesis risks becoming unfalsifiable if every post-hoc feature is called a representation. We therefore impose several constraints.

- A candidate representation must be specified independently enough to support preregistered probes or interventions; purely post-hoc naming is insufficient.
- Representation emergence must have temporal relevance to generalization. A feature that appears only after grokking cannot explain the transition.
- Correlation is not sufficient. At least one causal intervention should selectively alter the predicted capability.
- Explicitly providing the representation should change learning in the predicted direction if discovery cost is claimed to be important.
- Capacity-dependent representation claims require matched task/data/training controls across multiple seeds, not a two-model anecdote.
- The M04/M05 spectral rules are currently decision-level surrogates, not complete proofs of the internal algorithm.
- DLP at small smooth group orders does not imply cryptographic-scale DLP competence. Prime-order and larger-group controls are mandatory before broader claims.
- A failure to find the proposed representation while grokking remains robust across tasks would directly weaken the hypothesis and should be reported as such.

Study program required to test Representation-Relative Grokking

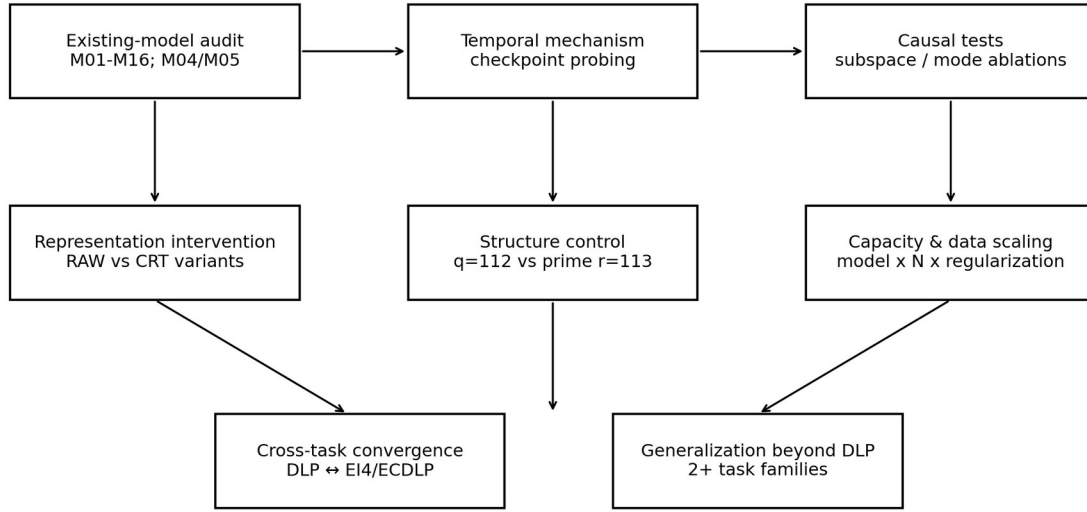


Figure 3. Experimental program required to move from a motivating hypothesis to a comprehensive study.

13. Plan to Complete the Study

A complete paper should not rely on M04 and M05 alone. The following staged program is designed to distinguish a publishable motivating result from a convincing general theory paper.

Phase A - Freeze the empirical foundation

- Regenerate M04 and M05 metrics directly from immutable checkpoints and split manifests; hash all artifacts.
- Rerun spectral extraction with conjugate-pair canonicalization and report spectral entropy, participation ratio, Gini, K90/K99, logit rank, and circulancy.
- Run residual-logit tests: sparse surrogate alone, residual alone, and targeted frequency removal.
- Extract the same metrics from the wider M01-M16 family wherever grokked checkpoints exist.

Phase B - Establish temporal precedence

- Save dense checkpoints around memorization and grokking for M04 and M05.
- Probe cyclic coordinates, divisor-aligned residues, and spectral subspaces across training.
- Define t_R for each candidate representation and test whether $t_R \leq t_{\text{grok}}$ reproducibly across seeds.
- Compare representation trajectories across model sizes, not only final checkpoints.

Phase C - Explicit representation interventions

- Run RAW, CRT-BOTH, CRT-16-ONLY, CRT-7-ONLY, RAW+CRT, and scrambled-pair controls on identical DLP examples.
- Measure grokking time, direct-generalization incidence, N_{crit} , and examples/tokens/FLOPs to generalization.
- Confirm single-component information ceilings and audit any violation for leakage.
- Test whether explicit CRT input collapses or shortens the delayed phase as predicted.

Phase D - Prime-order structure control

- Run the matched $q=112$ composite versus $r=113$ prime-order experiment with the exact M04 architecture.
- Use both coverage-matched and sample-count-matched splits plus Base-OOD.

- Perform the same spectral, probe, and causal analyses on prime-order checkpoints.
- Extend to at least 3-5 additional prime-order subgroup sizes to avoid a one-modulus conclusion.

Phase E - Capacity and data scaling

- Choose landmark sizes spanning roughly 100k, 425k, 657k, 4M, 16M, and 100M parameters where feasible.
- For each landmark, use matched data and at least 5 seeds near the grokking frontier.
- Test whether model size predicts representation density, discovery time, or route selection after controlling for training data.
- Fit exploratory scaling relationships only after inspecting whether monotonic or piecewise models are justified.

Phase F - Causal mechanism

- Perform rank-matched subspace ablations targeted at period-7, period-16, and alternative M05 modes.
- Test predicted double dissociations on $x \bmod 7$ versus $x \bmod 16$.
- Use activation patching or representation replacement between pre-grok and post-grok checkpoints where technically stable.
- Distill minimal decision circuits and compare them with causal necessity rather than only predictive fit.

Phase G - Cross-task convergence

- Compare multiplicative DLP with EI4/ECDLP checkpoints using common analysis-only cyclic coordinates.
- Train explicit-coordinate controls and quantify convergence using CKA/RSA/CCA/Procrustes/subspace angles.
- Run the pairing-mediated EC representation experiment if the underlying curve/torsion conditions are valid.
- Test whether representation similarity increases specifically at or before grokking.

Phase H - Generality beyond DLP

- Add at least two further mathematical task families with known alternative representations. Good candidates include modular roots with p-adic/character coordinates and matrix/subspace tasks with invariant-coordinate representations.
- For each task, predefine a surface representation, one candidate symmetry-adapted representation, an explicit-representation intervention, and a causal test.
- The paper should be willing to report tasks where the hypothesis fails or where a different mechanism dominates.

Phase I - Statistical and reproducibility standard

- Use two seeds only for discovery; use at least five for key claims and more around unstable phase boundaries.
- Report confidence intervals for grokking probability, t_{grok} , and representation metrics.
- Keep optimizer steps, unique examples, tokens, and approximate FLOPs as separate compute axes.
- Freeze code, environment, data generators, manifests, hashes, and checkpoint metadata before final analysis.

Phase J - Final paper package

- Lock the hypothesis and falsification criteria before the final confirmatory runs.
- Separate exploratory analyses from confirmatory figures.
- Prepare an anonymized code/data release with deterministic dataset generation and analysis scripts.
- Target the final narrative around representation discovery and causality, with DLP as the primary model system rather than as a cryptanalytic claim.

13.1 Evidence gates for a complete study

Gate	Minimum evidence	Purpose
G1	M04/M05 metrics regenerated from frozen	Remove report-level ambiguity

Gate	Minimum evidence	Purpose
	checkpoints	
G2	Representation signal precedes grokking across ≥ 5 seeds	Temporal relevance
G3	Explicit representation materially reduces discovery cost	Intervention/sufficiency
G4	Targeted ablation harms predicted component more than random ablation	Causal necessity
G5	Prime-order control either changes the mechanism or falsifies the smooth-order explanation	Structural specificity
G6	Capacity sweep reproduces representation-route differences	Generalize beyond M04/M05
G7	At least one cross-task convergence result or one strong negative result	Test latent-mathematics claim
G8	At least two non-DLP mathematical task families tested	Support broader hypothesis

13.2 Proposed final figure set

- Figure 1: formal representation-transition schematic and definitions.
- Figure 2: M01-M16 capacity versus grokking time versus spectral/circuit complexity.
- Figure 3: M04 and M05 checkpoint trajectories showing representation emergence before generalization.
- Figure 4: RAW versus explicit CRT representation learning curves and sample-complexity frontier.
- Figure 5: $q=112$ composite versus $r=113$ prime-order spectral/mechanistic comparison.
- Figure 6: causal ablations with component-specific double dissociation.
- Figure 7: accessibility-efficiency frontier across architectures, seeds, and model sizes.
- Figure 8: DLP versus EI4/ECDLP representation convergence, with explicit-coordinate controls.
- Figure 9: generalization of the framework to at least two additional mathematical task families.

14. Discussion and Outlook

The proposed theory reframes grokking on mathematical tasks as a search over representations as well as computations. M04 and M05 motivate this view because the same task admits different spectral routes with different apparent accessibility and complexity. The theory becomes scientifically useful only if those representations can be tracked before grokking, manipulated directly, and shown to matter causally. The planned CRT intervention, prime-order control, capacity sweep, and DLP-to-ECDLP convergence study are designed precisely for that purpose.

If the evidence supports the hypothesis, the broader implication is that neural mathematical reasoning may be governed by a representation search problem: the effective difficulty of a task depends on the cost of discovering coordinates in which the rule becomes simple. If the evidence does not support it, the same experimental program will still clarify which aspects of spectral structure are consequences rather than causes of grokking.

References

- [1] Power, A., Burda, Y., Edwards, H., Babuschkin, I., & Misra, V. (2022). Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets. arXiv:2201.02177.

- [2] Nanda, N., Chan, L., Lieberum, T., Smith, J., & Steinhardt, J. (2023). Progress Measures for Grokking via Mechanistic Interpretability. ICLR 2023; arXiv:2301.05217.
- [3] Chughtai, B., Chan, L., & Nanda, N. (2023). A Toy Model of Universality: Reverse Engineering How Networks Learn Group Operations. ICML 2023, PMLR 202:6243-6267; arXiv:2302.03025.
- [4] Nguyen, H. D. (2026). The Discrete-Log Clock: How a Transformer Learns Modular Multiplication. arXiv:2606.17399.
- [5] Varma, V., Shah, R., Kenton, Z., Kramar, J., & Kumar, R. (2023). Explaining Grokking through Circuit Efficiency. arXiv:2309.02390.
- [6] Kumar, T., Bordelon, B., Gershman, S., & Pehlevan, C. (2024). Grokking as the Transition from Lazy to Rich Training Dynamics. ICLR 2024.
- [7] Takhanov, R., Tezekbayev, M., Pak, A., Bolatov, A., Kadyrsizova, Z., & Assylbekov, Z. (2023). Intractability of Learning the Discrete Logarithm with Gradient-Based Methods. arXiv:2310.01611.

Appendix A. Operational Metrics Proposed for Confirmatory Work

Metric	Operational definition
t_mem	first checkpoint with $\geq 99\%$ training accuracy
t_10, t_50, t_90, t_99	first checkpoint crossing the corresponding held-out accuracy threshold
Delta t_grok	t_90 - t_mem for delayed-generalization runs
K99	minimum canonical spectral-mode set whose fitted decision rule reaches $\geq 99\%$ argmax agreement
Spectral participation ratio	effective number of energetic modes after conjugate-pair canonicalization
Representation alignment score	predeclared CKA/RSA/probe similarity to a candidate representation or explicit-coordinate control
Causal representation effect	difference in held-out/OOD performance under targeted versus rank-matched random ablation
N_crit	minimum unique training examples yielding reliable generalization under a fixed protocol
Direct-generalization flag	train and test performance rise together without a delayed phase; treated as a distinct success regime, not as failure to grok